

objects are stuck-in and trapped at the core and a high-intensity cage that envelopes an object and holds it in place from all directions.

RoboBee, a tiny robot that can fly, swim

Scientists have designed an insect-like robot, smaller than a paperclip, that can both fly and swim, paving the way for future dual aerial-aquatic robotic vehicles.

The biggest challenge was conflicting design requirements. Aerial vehicles require large airfoils like wings or sails to generate lift, while underwater vehicles need to minimise surface area to reduce drag. To solve this problem, engineers at Harvard University's John



First insect-size robot that can fly, swim
(Image courtesy: www.bravejournal.in)

A. Paulson School of Engineering and Applied Science (SEAS) took a clue from puffins. These birds employ similar flapping motions to propel themselves through air as through water.

According to Kevin Chen, a graduate student at Harvard Microrobotics Lab at SEAS, "Through various theoretical, computational and experimental studies, we found that the mechanics of flapping propulsion are actually very similar in air and in water."

Robot that uses artificial brain cells to navigate like a human

Behaviour and interplay of two types of neurons in the brain gives humans and other animals the ability to navigate by building a mental map of their surroundings. Now, a robot has been provided with a similar cluster of virtual cells to help it find its own way around. This work could help neuroscientists understand the functioning of the brain's navigation system.

Researchers in Singapore simulated two types of cells (place and grid cells) known to be used for navigation in the brain and enabled a small-wheeled robot to find its way around. Rather than simulate the cells physically, they created a 2D model of the cells in software.

Researchers had the robot roam around the office space, and verified that its artificial place and grid cells functioned in a comparable way to their biological counterparts.

LCD eye lens that could restore hyperopia

Devesh Mistry, a postgraduate research student at University of Leeds, is developing a new eye lens that could restore long-sightedness in older people. People above the age of 45 need reading glasses because they suffer from presbyopia, a condition that makes their eye

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